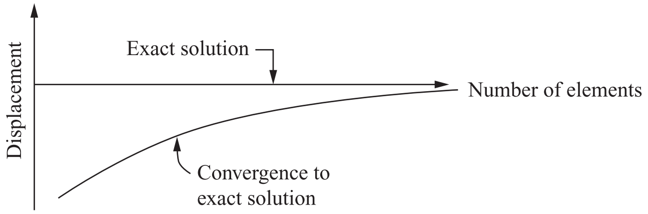

■ **Figure 3–6** Convergence to the exact solution for displacement as the number of elements of a finite element solution is increased

The idea that the interpolation (approximation) function must allow for a rigid-body displacement means that the function must be capable of yielding a constant value (say, a_1), because such a value can, in fact, occur. Therefore, we must consider the case

$$u = a_1 \tag{3.2.10}$$

For $u = a_1$ requires nodal displacements $u_1 = u_2$ to obtain a rigid-body displacement. Therefore

$$a_1 = u_1 = u_2 \tag{3.2.11}$$

Using Eq. (3.2.11) in Eq. (3.2.8), we have

$$u = N_1 u_1 + N_2 u_2 = (N_1 + N_2) a_1 \tag{3.2.12}$$

From Eqs. (3.2.10) and (3.2.12), we then have

$$u = a_1 = (N_1 + N_2) a_1 \tag{3.2.13}$$

Therefore, by Eq. (3.2.13), we obtain

$$N_1 + N_2 = 1 \tag{3.2.14}$$

Thus Eq. (3.2.14) shows that the displacement interpolation functions must add to unity at every point within the element so that u will yield a constant value when a rigid-body displacement occurs.

3.3 Transformation of Vectors in Two Dimensions

In many problems it is convenient to introduce both local $(x' - y')$ and global (or reference) $(x - y)$ coordinates. Local coordinates are always chosen to represent the individual element conveniently. Global coordinates are chosen to be convenient for the whole structure.

Given the nodal displacement of an element, represented by the vector **d** in Figure 3–7, we want to relate the components of this vector in one coordinate system to components in another. For general purposes, we will assume in this section that **d** is not coincident with either the local or the global axis. In this case, we want to relate global displacement components to local ones. In so doing, we will develop a transformation matrix that will subsequently be used to develop the global stiffness matrix for a bar element. We define the angle θ to be positive when measured counterclockwise from x to x'. We can express vector displacement **d** in both global and local coordinates by

$$\mathbf{d} = u\mathbf{i} + v\mathbf{j} = u'\mathbf{i}' + v'\mathbf{j}' \tag{3.3.1}$$

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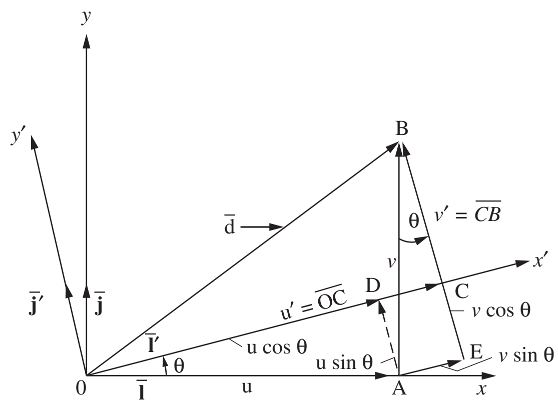

■ **Figure 3–7** General displacement vector **d** in two dimensions

where **i** and **j** are unit vectors in the x and y global directions and **i**′ and **j**′ are unit vectors in the x' and y' local directions. From Figure 3–7, we have the following vectors in terms of reference letters as

$$\bar{u} = \overline{OA},\ \bar{v} = \overline{AB},\ \bar{u}' = \overline{OC},\ \bar{v}' = \overline{CB} \tag{3.3.2}$$

From Figure 3–7, we can observe by vector addition along the x' axis, the relationship

$$\overline{OC} = \overline{OD} + \overline{DC} \tag{3.3.3}$$

Using standard trigonometric relations in Figure 3–7 and use of Eq. (3.3.2), we obtain

$$\overline{OD} = \overline{OA}\cos\theta = \bar{u}\cos\theta \text{ and } \overline{DC} = \overline{AE} = \bar{v}\sin\theta \tag{3.3.4}$$

Using Eq. (3.3.2) for u' and Eq. (3.3.4) in Eq. (3.3.3), we have

$$u' = u\cos\theta + v\sin\theta \tag{3.3.5}$$

In a similar fashion, by vector addition in the y' direction of Figure 3–7, we have

$$\overline{CB} = -\overline{AD} + \overline{BE} \tag{3.3.6}$$

Again using standard trigonometric relations in Figure 3–7 and use of Eq. (3.3.2), we have

$$\overline{AD} = \overline{OA}\sin\theta = \bar{u}\sin\theta \text{ and } \overline{BE} = \overline{AB}\ \cos\theta = \bar{v}\cos\theta \tag{3.3.7}$$

Now using Eq. (3.3.2) for v' and Eq. (3.3.7) in Eq. (3.3.6), we obtain

$$v' = -u\sin\theta + v\cos\theta \tag{3.3.8}$$

Expressing Eqs. (3.3.5) and (3.3.8) together in matrix form, we get

$$\begin{Bmatrix} u' \\ v' \end{Bmatrix} = \begin{bmatrix} C & S \\ -S & C \end{bmatrix} \begin{Bmatrix} u \\ v \end{Bmatrix} \tag{3.3.9}$$

where $C = \cos\theta$ and $S = \sin\theta$.

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